

Clarification of ‘Algorithmic Collusion without Threats’

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Abstract

This brief note clarifies that the scenario described in Arunachaleswaran et al. (2025)—titled ‘Algorithmic Collusion without Threats’—is not one of collusion, but one where one player is behaving non-competitively and the other is behaving competitively.

It is important for academic discourse at the interface between computer science and law to identify definitions for concepts that make sense both in the legal domain and in the computational domain. *Algorithmic collusion*—where there is a concern that algorithms might facilitate the coordination of sellers on supra-competitive prices—is one such concept. This brief note is to point out that the phenomenon studied by Arunachaleswaran et al. (2025) is not algorithmic collusion.

Arunachaleswaran et al. (2025)—titled ‘Algorithmic Collusion without Threats’—shows that very simple strategies sustain supra-competitive prices in Nash equilibrium of a simple repeated pricing game. In this equilibrium, one player plays by a static price distribution and the other player plays a strategy that learns to best respond (at least) to any static price distribution. In this Nash equilibrium of the repeated game, both players obtain a constant fraction of the monopoly revenue, while the “competitive” Nash equilibrium of the stage game has (approximately) no revenue for either player. The authors claim that both players’ strategies are reasonable in that they are optimizing against their opponent’s strategy. The second player is reasonable for learning to best respond to the induced distribution of the demand. Given that the second player is doing this, the first player is reasonable in responding to the second player in a way that improves revenue.

While the outcome above is non-competitive, it is not a collusion. Collusion fundamentally requires both players to be complicit. Only the first player in this outcome is behaving non-competitively. In fact, the informal argument that the first player is behaving reasonably is flawed.

There is an important distinction in the study of repeated games between equilibria of the stage game that are repeated and equilibria of the repeated game as a whole. Stage game equilibria are generally a subset of a much larger set of repeated game equilibria. The economics literature is unambiguous that *competitive prices* are ones that result from an equilibrium in the stage game. Equilibria of a repeated game that are not equilibria of the stage game are, by this definition, *non-competitive*. The folk theorem suggests that there are many such non-competitive equilibria. Some are from strategies with threats like tit-for-tat and some—as obtained in Arunachaleswaran et al. (2025)—are not. The former looks like collusion; the latter does not, but both are non-competitive. For both cases, strategic optimization at the level of the repeated game is exactly the “unreasonable” behavior that needs to be curtailed to enable competitive prices.

Collusion is a special case of non-competitive behavior where, in the legal theory, agreements are made to distort prices away from competitive prices. Recently, the algorithmic collusion literature has also considered tacit collusion where, rather than sellers making an agreement, the algorithms learn dynamics that sustain supra-competitive prices. In all of the legal and academic discourse on collusion, it is fundamentally about the choices of two or more parties that result in non-competitive prices. An outcome can only be collusion if choices of both players are implicable in arriving at that outcome. Specifically, in a collusion there must always be a reasonable strategy that a player can make that is non-collusive.

Algorithms are used to price because market conditions such as the demand and level of competitive prices are unknown. A minimum requirement to consider a pricing algorithm as good is, given a static demand, i.e., the probability of sale at every price is the same across rounds, the pricing algorithm learns to optimize with respect to this demand. There may be more requirements of good algorithms when considering

dynamic settings of price competition, e.g., see Hartline (2026), but these additional requirements cannot be distinguished when considering only static demand. Thus, for static demand an algorithm is indistinguishable from a good algorithm if it learns to best respond.

As any reasonable definition of collusion requires that there be a strategy that is non-collusive, and as algorithms that learn to best respond against static demand are indistinguishable from good algorithms, an algorithmically reasonable definition of collusion cannot call an outcome collusion when all but one player is best responding to a static induced demand.

While Arunachaleswaran et al. (2025) is not a paper about collusion, it does have an important message for the literature on pricing algorithms. A regulator may struggle to distinguish between sellers who naïvely price without optimizing and sellers who are manipulating the market, like the first player in their example. The lesson is that to ensure competitive prices, both such sellers must be regulated against. Such regulation may be difficult in practice.

References

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